1. Case Study: Library Management System using Inheritance in Java

Problem Scenario:

A college library needs a system to manage different types of books and resources. All resources (Books, Magazines, and Reference Books) share common properties such as **title, author, and publication year**. However, each resource also has some **special characteristics**:

- **Book**: Has a genre and numberOfPages.
- **Magazine**: Has an issueNumber and monthOfPublication.
- **ReferenceBook**: Has a subject and can only be read within the library (not issued).

The library management system should use **Inheritance** to avoid code duplication and ensure reusability.

Class Design:

1. **Parent Class – LibraryResource**
    - Attributes: title, author, publicationYear
    - Methods: displayDetails()
2. **Child Classes**
    - **Book**: Inherits LibraryResource, adds genre, numberOfPages.
    - **Magazine**: Inherits LibraryResource, adds issueNumber, monthOfPublication.
    - **ReferenceBook**: Inherits LibraryResource, adds subject.

2. Case Study: Employee Management System using Inheritance in Java

Problem Scenario

A company employs different categories of employees. All employees share common details such as **employee ID, name, and salary**. However, each category of employee has some special features:

- **Manager**: Has a department and can approve leave.
- **Developer**: Has a programming language specialization and can work on projects.
- **Intern**: Has a duration (months) and a mentor name.

The system should use **Inheritance** to avoid duplicate code and to represent relationships clearly.

Class Design

1. **Parent Class – Employee**
    - Attributes: empId, name, salary
    - Methods: displayDetails()
2. **Child Classes**
    - **Manager**: Inherits Employee, adds department.
    - **Developer**: Inherits Employee, adds programmingLanguage.
    - **Intern**: Inherits Employee, adds duration, mentorName.